

Introduction to Analytics

Complete Detailed Notes for Data Science, Data Analytics, Analytics Life Cycle, QA, Deployment, and Real-World Examples

Prepared from the uploaded class notes: Introduction_to_Analytics.pdf

1. What is Data?

Data means a collection of facts. It may be collected from observations, experiments, transactions, sensors, forms, websites, images, text, audio, or video. Data is the raw material from which information and knowledge are created.

Type of Data	Examples
Numbers	Marks, salary, age, sales, temperature
Text	Reviews, names, comments, doctor notes
Images	X-rays, product photos, scanned documents
Audio/Video	Call recordings, CCTV footage, lectures

Example: A hospital may collect patient age, blood pressure, lab reports, doctor notes, treatment history, and X-ray images. All of these are forms of data.

2. What is Data Science?

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, machine learning, and domain knowledge to collect, process, analyze, and interpret large amounts of data. The aim is to extract meaningful insights and support decision-making.

Data Science = Raw Data -> Useful Information -> Prediction or Decision

In simple words, data science converts raw data into useful information, predictions, and business actions.

- Netflix recommends movies using viewing history.
- Amazon recommends products using browsing and purchase behavior.
- Banks detect fraud using transaction patterns.
- Hospitals predict disease risk using patient records.

3. Why Data Science is Important

Organizations generate huge amounts of data every day. Without data science, this data remains unused. With data science, businesses can understand customers, reduce cost, increase sales, detect risks, and automate decisions.

- Improves decision-making with evidence instead of guesswork.
- Finds hidden patterns in large datasets.
- Predicts future outcomes such as demand, churn, fraud, and disease risk.

- Automates repetitive or complex decisions using machine learning models.
- Improves customer experience through personalization.

4. Components of Data Science

Data science is not only model building. It includes the complete flow from collecting data to deploying and maintaining models.

Component	Meaning	Example
Data Collection	Gathering data from many sources	Databases, APIs, sensors, Excel, logs
Data Cleaning	Fixing errors and improving quality	Missing values, duplicates, wrong formats
Exploratory Data Analysis	Understanding patterns and relationships	Mean, correlation, charts, trends
Data Modeling	Building statistical or ML models	Fraud detection, churn prediction
Data Visualization	Presenting insights visually	Power BI, Tableau, charts, dashboards
Deployment & Maintenance	Putting model into real use and monitoring	Using fraud model in production

4.1 Data Collection

Data collection means gathering data from different sources. A business may collect data from databases, websites, APIs, mobile apps, IoT devices, social media, surveys, spreadsheets, and logs.

Example: A hospital collects patient records, lab test results, treatment history, doctor notes, and billing data.

4.2 Data Cleaning

Raw data is usually messy. It may have missing values, duplicate records, spelling mistakes, invalid numbers, inconsistent formats, and outliers. Data cleaning fixes these problems before analysis or modeling.

Raw Data Problem	Example	Possible Fix
Missing value	Age is blank	Fill with median or ask source system
Duplicate record	Same patient ID repeated	Remove duplicate
Invalid value	Age = 999	Correct or remove
Wrong format	Date as 12-05-26 and 2026/05/12	Standardize format

4.3 Exploratory Data Analysis (EDA)

EDA means exploring the dataset before building a model. It helps identify patterns, trends, relationships, outliers, and errors.

- What is the average sale?
- Which product sells most?
- Are there missing values or outliers?
- Is price related to sales?

- Which customer group buys more?

4.4 Data Modeling

Data modeling means using statistics or machine learning algorithms to make predictions or decisions. The model learns patterns from past data and applies them to new data.

Problem	Model Type
Predict house price	Regression
Predict pass/fail	Classification
Group similar customers	Clustering
Detect unusual transaction	Anomaly detection / classification

4.5 Data Visualization

Data visualization presents data using charts, graphs, and dashboards. Visuals make insights easier to understand than raw rows and columns.

- Bar chart for category comparison.
- Line chart for trends over time.
- Pie chart for percentage share.
- Scatter plot for relationship between two variables.
- Dashboard for business monitoring.

4.6 Deployment and Maintenance

Deployment means putting a model or analytical solution into real use. Maintenance means monitoring and improving it over time.

Example: A fraud detection model is deployed in a banking system. Every transaction is checked by the model. If the data pattern changes, the model may need retraining.

5. Data Science Life Cycle

The data science life cycle is a structured process that moves from raw data to useful decisions.

Collection -> Cleaning -> Exploratory Data Analysis -> Model Building -> Model Deployment

Example: For a customer churn project, the team collects customer data, cleans missing values, studies customer behavior, builds a churn prediction model, and deploys it in the CRM system.

6. Prerequisites for Data Science

Prerequisite	Why It Is Needed
Machine Learning	To build prediction and classification models
Modeling	To select the correct algorithm for the problem
Statistics	To understand data, probability, correlation, and inference

Programming	To clean, analyze, and model data using Python or R
Database	To store, query, and extract data using SQL or other tools

Important Python libraries include Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PySpark, TensorFlow, and Jupyter Notebook.

7. Data Science Roles

Role	Main Work	Example
Data Engineer	Builds pipelines to collect, move, clean, and store data	Daily sales data pipeline
Data Scientist	Builds models and extracts actionable insights	Churn prediction model
MLOps Engineer	Deploys and monitors ML models in production	Fraud model deployment
BI Analyst	Builds reports and dashboards for business leaders	Sales leaderboard
Data Analyst	Analyzes existing data to answer business questions	Why sales dropped last month
Data Product Manager	Manages data products and ensures business value	Business intelligence engine roadmap

These roles are connected. Data engineers prepare clean data. Analysts and BI teams report insights. Data scientists build models. MLOps engineers deploy models. Product managers ensure the solution delivers business value.

8. Data Science Tools

Purpose	Tools
Data Analysis	SAS, Jupyter, RStudio, MATLAB, Excel
Data Warehousing	Informatica, Talend, AWS Redshift
Visualization / ML	Tableau, Power BI, Spark MLlib, Mahout, Azure ML Studio
Programming	Python, R, SQL
Big Data	Hadoop, Spark, Hive, Kafka

9. What is Data Analytics?

Data analytics is the process of examining data to uncover useful insights, patterns, and trends that help in decision-making.

It answers business questions such as:

- Why are sales decreasing?
- Which product is most profitable?
- Which customers are inactive?
- Which region has the highest demand?

10. Data Science vs Data Analytics

Data Science	Data Analytics
Broader field	More focused field
Uses ML, AI, statistics, programming, and domain knowledge	Knowledge in analyzing data and finding insights
Predicts future and automates decisions	Explains past and present
Builds models	Builds reports, dashboards, and insights
Example: customer churn prediction model	Example: monthly sales report

Simple memory: Data Analytics = Understand data. Data Science = Understand + Predict + Automate decisions.

11. Types of Analytics

Type	Question Answered	Meaning	Example
Descriptive Analytics	What happened?	Summarizes historical data	Monthly sales report
Diagnostic Analytics	Why did it happen?	Finds reasons behind events	Sales dropped due to high shipping costs
Predictive Analytics	What is likely to happen?	Forecasts future outcomes	Next month demand prediction
Prescriptive Analytics	What should be done?	Recommends actions	Offer discount or free shipping

11.1 Descriptive Analytics

Descriptive analytics summarizes past data. It tells what happened. Examples include monthly sales reports, customer demographics, attendance summaries, and website traffic reports.

11.2 Diagnostic Analytics

Diagnostic analytics investigates why something happened. If sales dropped, it checks causes such as price changes, poor reviews, competitor discounts, supply issues, or website problems.

11.3 Predictive Analytics

Predictive analytics uses statistics and machine learning to estimate what may happen in the future. Examples include sales forecasting, churn prediction, loan default prediction, and disease risk prediction.

11.4 Prescriptive Analytics

Prescriptive analytics recommends what action should be taken. Examples include recommending discounts, optimizing prices, assigning delivery routes, or scheduling hospital staff.

12. Easy Example of Four Analytics Types

Suppose an online store has decreasing sales.

Type	Question	Answer Example
Descriptive	What happened?	Sales decreased by 20% last month
Diagnostic	Why did it happen?	Many customers left at checkout because shipping cost was high

Predictive	What will happen?	Sales may drop again next month if shipping cost remains high
Prescriptive	What should be done?	Introduce free shipping above Rs. 999

13. Data Analytics Life Cycle

The data analytics life cycle is a step-by-step process used to transform raw data into meaningful insights and business action. It is especially useful for big data and data science projects.

Discovery -> Data Preparation -> Model Planning -> Model Building -> Communicate Results -> Operationalize

13.1 Phase 1: Discovery

Discovery is the first phase. Here, the business problem is defined and understood before touching the data.

- Define the business problem or objective.
- Identify the scope of the project.
- Determine relevant data sources.
- Frame initial hypotheses.
- Discuss with stakeholders and understand requirements.
- Assess available resources, tools, data, and team expertise.

Example: A hospital wants to reduce patient waiting time. The discovery phase defines the problem, objective, stakeholders, and required data.

13.2 Phase 2: Data Preparation

Data preparation is often the most time-consuming phase because poor data leads to poor results.

- Collect data from databases, APIs, spreadsheets, or logs.
- Perform ETL: Extract, Transform, Load.
- Remove duplicates and handle missing values.
- Correct inconsistencies and invalid records.
- Standardize and normalize data where required.
- Create useful features for analysis or modeling.

Example: Cleaning stock market data by fixing dates, removing duplicate rows, and preparing variables for investment analysis.

13.3 Phase 3: Model Planning

In model planning, analysts decide how to analyze the data and which modeling technique is suitable.

- Define problem statements and objectives.
- Select relevant variables or features.
- Choose algorithms such as regression, classification, or clustering.

- Consider model complexity, interpretability, and scalability.
- Use EDA techniques like correlation matrices and scatter plots.

Example: Planning a machine learning model for customer churn prediction.

13.4 Phase 4: Model Building

In model building, the model is actually created, trained, tested, and improved.

- Split data into training and testing datasets.
- Build and train models using selected algorithms.
- Evaluate model using appropriate metrics.
- Fine-tune parameters for better accuracy.
- Use cross-validation techniques such as K-fold validation.
- Avoid overfitting by testing on unseen data.

Overfitting means the model performs well on training data but poorly on new data. It is like a student memorizing only previous questions but failing on new ones.

13.5 Phase 5: Communicate Results

Analysis is only useful when results are communicated clearly. Stakeholders may not understand technical details, so insights must be presented in business language.

- Create reports, dashboards, and visualizations.
- Document insights and recommendations.
- Explain findings clearly to stakeholders.
- Connect results with business objectives.

Example: Instead of saying "model coefficient is significant", say "high shipping charges are causing customers to abandon carts".

13.6 Phase 6: Operationalize

Operationalize means deploying the final solution into real business use and monitoring it continuously.

- Deploy the model into production.
- Integrate it with business processes or applications.
- Automate predictions or decisions.
- Monitor performance and data quality.
- Update or retrain the model when data changes.
- Document deployment and maintenance processes.

Example: Deploying a fraud detection model in a banking system.

14. E-commerce Analytics Example

The notes give a practical e-commerce example where sales are decreasing.

Step	Action	Example
Problem	Define the issue	Sales are decreasing
Collect Data	Gather relevant data	Website visits, orders, customer behavior
Explore	Find patterns	High cart abandonment
Clean Data	Fix data quality issues	Remove errors, fill missing values
Analyze	Identify cause	High shipping cost
Visualize	Show findings	Charts showing checkout drop-off
Act	Take business action	Free shipping above Rs. 999
Monitor	Check results	Sales increase or re-analyze

This cycle is not one-time. Every new decision creates new data, which leads to better insights over time.

15. Quality Assurance and Documentation

Quality Assurance ensures that the model and results are accurate, reliable, unbiased, and ready for real-world use.

- Test the model for accuracy, reliability, bias, and edge cases.
- Check whether the model performs correctly before deployment.
- Evaluate using accuracy, precision, recall, and F1-score.
- Verify that insights are logical and trustworthy.

Documentation records the full process: data sources, cleaning steps, assumptions, methods, models, metrics, results, limitations, and future improvements.

Good documentation ensures reproducibility, transparency, and smooth knowledge transfer to other teams.

Metric	Meaning
Accuracy	Overall correct predictions out of all predictions
Precision	Out of predicted positives, how many were truly positive
Recall	Out of actual positives, how many were correctly found
F1-score	Balance between precision and recall

16. Management Approval and Installation

Before deploying a model, stakeholders must approve it. They review business value, ROI, risks, compliance, data privacy, and strategic alignment.

ROI means Return on Investment. It checks whether the expected benefits are greater than the cost of the project.

Installation means placing the approved system into a live environment. It includes API integration, infrastructure setup, monitoring tools, logging, access control, and rollback plans.

A rollback plan defines what to do if deployment fails or the model behaves incorrectly.

17. Acceptance and Operations

After installation, the system must be accepted by users and operated continuously.

- User Acceptance Testing checks whether outputs are correct and usable.
- Users are trained to interpret dashboards or model outputs.
- Teams track model performance, data quality, drift, and system uptime.
- SLAs, feedback loops, and operational procedures are created.
- Models and pipelines are updated as business conditions change.

18. Intelligent Data Analysis

Intelligent Data Analysis is the continuous phase where AI and ML techniques are used for automated insights, pattern recognition, and adaptive learning.

- The system becomes smarter over time using new data.
- Models are monitored and improved regularly.
- Insights are presented through charts, dashboards, and reports.
- Recommendations are aligned with business goals.
- Comprehensive reports document the full data analysis process.

Example: A recommendation system improves as users keep clicking, purchasing, and rating products.

19. Life Cycle of Data Analysis in Hospital

The PDF gives a hospital example showing how analytics works in healthcare.

Stage	Meaning	Hospital Example
Data Collection	Gather relevant data	Patient records, lab results, treatment history, doctor notes
Data Cleaning	Remove errors and inconsistencies	Correct duplicate patient IDs and invalid timestamps
Exploration & Analysis	Find patterns and trends	Identify peak admission hours and common illnesses
Data Modeling	Use statistics or ML for prediction	Predict high-risk diabetes patients
Visualization & Interpretation	Present insights clearly	Power BI dashboard for disease by age and region
Decision-Making & Action	Use insights for action	Allocate more beds, staff, and equipment to high-risk departments

20. Importance of Analytics

- Enhances decision-making.
- Improves business efficiency and competitiveness.
- Helps identify new opportunities.
- Helps reduce or mitigate risks.

- Provides personalized experiences for customers.

21. Complete Example: Analytics in a Shopping App

Suppose a shopping app has falling sales.

Step 1 - Discovery: Define the problem. Sales are decreasing. Objective: find the reason and increase sales.

Step 2 - Data Preparation: Collect website visits, cart data, orders, payment data, and customer reviews. Remove duplicate orders, fix missing customer IDs, and correct date formats.

Step 3 - Model Planning: Decide to use descriptive analytics for sales trends, diagnostic analytics to find causes, predictive analytics to forecast future sales, and prescriptive analytics to recommend actions.

Step 4 - Analysis / Model Building: Find that many users add products to cart but leave during checkout because shipping cost is high.

Step 5 - Communicate Results: Create a dashboard showing high drop-off at the shipping page and recommend free shipping above Rs. 999.

Step 6 - Operationalize: Apply free shipping rule, monitor sales, and re-analyze if results do not improve.

22. Interview-Level Differences

Concept	Meaning
Data	Raw facts such as numbers, text, images, or records
Data Science	Converts data into insights, predictions, and decisions
Data Analytics	Finds patterns, trends, and insights from data
Descriptive Analytics	Answers: What happened?
Diagnostic Analytics	Answers: Why did it happen?
Predictive Analytics	Answers: What will happen?
Prescriptive Analytics	Answers: What should be done?
Data Cleaning	Fixing errors, missing values, and inconsistencies
EDA	Exploring patterns and relationships
Model Building	Training and testing algorithms
Deployment	Putting model into real business use
QA	Testing accuracy, reliability, and bias
Documentation	Recording process, assumptions, and results

23. Final Summary

This chapter introduces analytics and data science. Data is raw facts. Data science converts raw data into useful information, predictions, and decisions. Data analytics studies data to find

patterns, trends, and insights.

The four major types of analytics are descriptive, diagnostic, predictive, and prescriptive. The analytics life cycle includes discovery, data preparation, model planning, model building, communication of results, and operationalization.

For real-world success, analytics projects also need quality assurance, documentation, management approval, installation, user acceptance, operations, and continuous intelligent analysis.

This topic is foundational for data science, machine learning, business intelligence, big data analytics, and data-driven decision-making.

24. Quick Revision Sheet

Question	Answer
What is data?	Raw facts collected from observations or experiments
What is data science?	Turning raw data into useful information and predictions
What is analytics?	Examining data to find insights, patterns, and trends
What is EDA?	Exploratory Data Analysis: understanding data before modeling
What is deployment?	Putting a model or dashboard into real-world use
What is QA?	Testing model and results for reliability
What is ROI?	Return on Investment: benefit compared to cost
What is UAT?	User Acceptance Testing by stakeholders or end-users